Supplementary file

Table of Contents

FIGURE S1. FUNNEL PLOT. OUTCOME: DIABETES PREVALENCE IN INPATIENT COVID-19 PATIENTS. 2

FIGURE S2. FUNNEL PLOT. OUTCOME: DEATHS IN HOSPITALIZED DIABETES PATIENTS WITH COVID-19 INFECTION. 2

FIGURE S3. FUNNEL PLOT. OUTCOME: DEATHS IN HOSPITALIZED COVID-19 PATIENTS WITH NO DIABETES. .. 3

TABLE S1. PREFERRED REPORTING ITEMS FOR SYSTEMATIC REVIEWS AND META-ANALYSES CHECKLIST[1]..... 4

TABLE S2. SUBGROUP PREVALENCE OF DEATH AMONG HOSPITALIZED COVID-19 PATIENTS WITH DIABETES MELLITUS BY CONTINENT. 7

TABLE S3. SUBGROUP PREVALENCE OF DEATH AMONG HOSPITALIZED COVID-19 PATIENTS WITH DIABETES MELLITUS BY ADMITTANCE TO THE INTENSIVE CARE UNIT 9

TABLE S4. UNIVARIATE META-REGRESSION ANALYSIS FOR THE PREVALENCE STUDIES ON DIABETES MELLITUS AMONG COVID-19 PATIENTS.[A,B] 12

TABLE S5. UNIVARIATE AND MULTIVARIATE META-REGRESSION ANALYSIS FOR THE PREVALENCE STUDIES ON DEATHS AMONG COVID-19 PATIENTS WITH DIABETES MELLITUS 13

TABLE S6. UNIVARIATE AND MULTIVARIATE META-REGRESSION ANALYSIS FOR THE PREVALENCE STUDIES ON DEATHS AMONG COVID-19 PATIENTS WITHOUT DIABETES MELLITUS. 14

TABLE S7. QUALITY ASSESSMENT OF THE REVIEWED STUDIES[*] 15

REFERENCE.....18

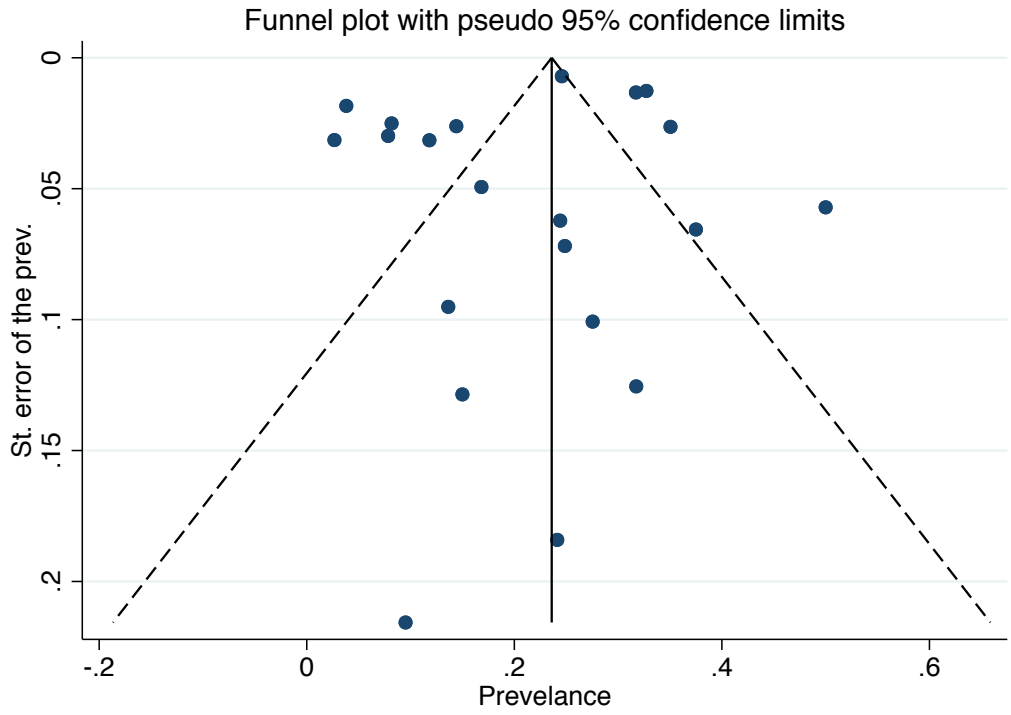


Figure S1. Funnel plot. Outcome: Diabetes prevalence in inpatient COVID-19 patients.

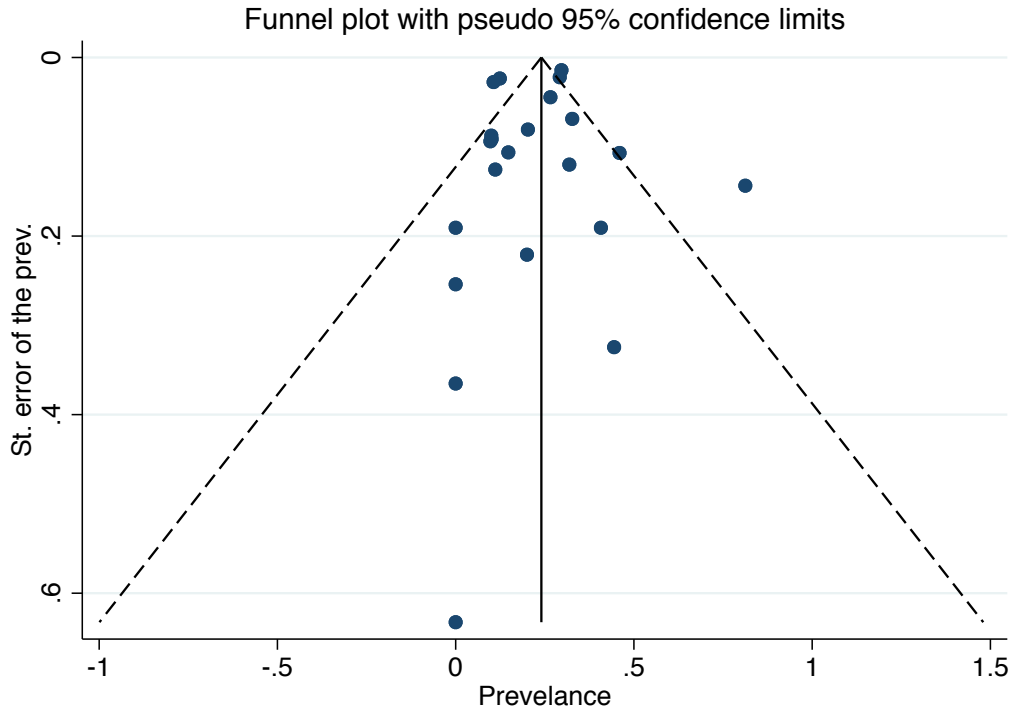


Figure S2. Funnel plot. Outcome: Deaths in hospitalized diabetes patients with COVID-19 infection.

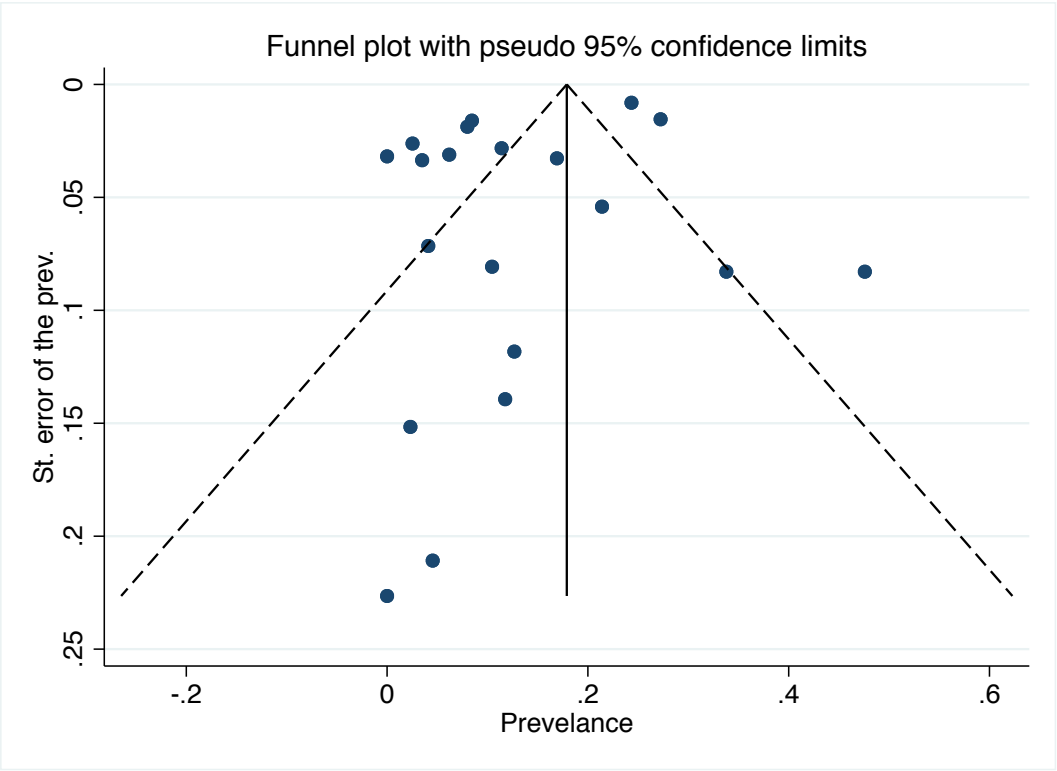


Figure S3. Funnel plot. Outcome: Deaths in hospitalized COVID-19 patients with no diabetes.

Table S1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist[1]

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5-6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6

Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6-7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7-8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7-8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	12
FUNDING			

Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	12

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

Table S2. Subgroup prevalence of death among hospitalized COVID-19 patients with diabetes mellitus by continent.

Diabetic COVID-19 patients in Asia*

Subgroup	Category	Number of studies	Number of DM patients	Number of deaths	Mean prevalence of deaths		95% prediction interval	Heterogeneity measures	
					%	95% CI		I^2 (%)	Q (p-value)
Country	China	9	767	175	20.0	8.0-34.0	0.0-76.0	94.35	<0.01
	India	1	2	0	0.0	0.0-84.0	Inestimable	-	-
	Iran	1	113	11	10.0	5.0-17.0	Inestimable	-	-
	Korea	2	42	11	21.0	9.0-35.0	Inestimable	-	-
	Oman	1	20	4	20.0	6.0-44.0	Inestimable	-	-
COVID-19 diagnosis	RT-PCR	12	783	151	13.0	6.0-21.0	0.0-47.0	84.64	<0.01
	Multiple modes	2	161	50	28.0	21.0-35.0	Inestimable	-	-
Diabetes diagnosis	Method specified	2	216	38	17.0	13.0-23.0	Inestimable	-	-
	Unclear	12	728	163	17.0	6.0-31.0	0.0-76.0	93.17	<0.01
DM type	Mixed T1 and T2 DM	-	-	-	-	-	-	-	-
	T2DM	2	55	39	71.0	58.0-83.0	Inestimable	-	-
	Unclear	12	889	162	13.0	7.0-21.0	0.0-0.46	85.39	<0.01
COVID-19 severity	Critically ill	3	64	43	40.0	0.0-93.0	Inestimable	-	-
	Not Critically ill	3	161	12	3.0	0.0-12.0	Inestimable	-	-
	Unclear	8	719	146	17.0	9.0-26.0	0.0-51.0	84.87	<0.01
Sample size	<100	9	218	65	17.0	1.0-44.0	0.0-100	93.02	<0.01
	≥100	5	726	136	16.0	8.0-26.0	0.0-59.0	90.85	<0.01
Overall		14	944	201	17.0	8.0-28.0	0.0-66.0	92.10	<0.01

Diabetic COVID-19 patients in Europe

Subgroup	Category	Number of studies	Number of DM patients	Number of deaths	Mean prevalence of deaths		95% prediction interval	Heterogeneity measures	
					%	95% CI		I^2 (%)	Q (p-value)
Country	France	1	1317	140	11.0	9.0-12.0	Inestimable	-	-
	Italy	1	69	22	32.0	21.0-44.0	Inestimable	-	-
	UK	2	5036	1509	30.0	29.0-31.0	Inestimable	-	-
COVID-19 diagnosis	RT-PCR	1	87	40	46	35.0-57.0	Inestimable	-	-
	Multiple modes	3	6335	1631	23	9.0-41.0	Inestimable	-	-
Diabetes diagnosis	Method specified	2	1404	180	12	10-14	Inestimable	-	-
	Unclear	2	5018	1491	30.0	28.0-31.0	Inestimable	-	-

DM type	Mixed T1 and T2 DM	3	1473	202	28.0	7.0-55.0	Inestimable	-	-
	T2DM	-	-	-	-	-	-	-	-
	Unclear	1	4949	1469	30.0	28.0-31.0	Inestimable	-	-
COVID-19 severity	Critically ill	-	-	-	-	-	-	-	-
	Not Critically ill	-	-	-	-	-	-	-	-
	Unclear	4	6422	1671	28.0	14.0-44.0	Inestimable	-	-
Sample size	<100	2	156	62	40.0	32.0-47.0	Inestimable	-	-
	≥100	2	6266	1609	25.0	24.0-26.0	Inestimable	-	-
Overall		4	6422	1671	28.0	14.0-44.0	0.0-96.0	98.87	<0.01

Diabetic COVID-19 patients in North America

Subgroup	Category	Number of studies	Number of DM patients	Number of deaths	Mean prevalence of deaths %	Mean prevalence of deaths 95% CI	95% prediction interval	Heterogeneity measures I^2 (%)	Heterogeneity measures Q (p-value)
Country	US	4	4445	968	20.0	11.0-32.0	0.0-79.0	98.35	<0.01
COVID-19 diagnosis	RT-PCR	4	4445	968	20.0	11.0-32.0	0.0-79.0	98.35	<0.01
	Multiple modes	-	-	-	-	-	-	-	-
Diabetes diagnosis	Method specified	1	88	13	15.0	8.0-24.0	Inestimable	-	-
	Unclear	3	4357	955	22.0	11.0-36.0	Inestimable	-	-
DM type	Mixed T1 and T2 DM	-	-	-	-	-	-	-	-
	T2DM	1	1808	224	12.0	11.0-14.0	-	-	-
	Unclear	3	2637	744	25.0	20.0-31.0	-	-	-
COVID-19 severity	Critically ill	-	-	-	-	-	-	-	-
	Not Critically ill	-	-	-	-	-	-	-	-
	Unclear	4	4445	968	20.0	11.0-32.0	0.0-79.0	98.35	<0.01
Sample size	<100	1	88	13	15.0	8.0-24.0	-	-	-
	≥100	3	4357	955	22.0	11.0-36.0	-	-	-
Overall		4	4445	968	20.0	11.0-32.0	0.0-79.0	98.35	<0.01

*For one study the deaths in the diabetes patients was based on the number of death data available for some of the diabetes patients[2]

Table S3. Subgroup prevalence of death among hospitalized COVID-19 patients with diabetes mellitus by admittance to the intensive care unit

Death in diabetic COVID-19 patients requiring ICU support

Subgroup	Category	Number of studies	Number of DM patients	Number of deaths	Mean prevalence of deaths		95% prediction interval	Heterogeneity measures	
					%	95% CI		I^2 (%)	Q (p-value)
Continent	Asia	2	201	70	34.0	27.0-41.0	Inestimable	-	-
	Europe	1	1317	140	11.0	9.0-12.0	Inestimable	-	-
	North America	1	1808	224	12.0	11.0-14.0	Inestimable	-	-
Country	China	2	201	70	34.0	27.0-41.0	Inestimable	-	-
	France	1	1317	140	11.0	9.0-12.0	Inestimable	-	-
	India	-	-	-	-	-	-	-	-
	Iran	-	-	-	-	-	-	-	-
	Italy	-	-	-	-	-	-	-	-
	Korea	-	-	-	-	-	-	-	-
	Oman	-	-	-	-	-	-	-	-
	UK	-	-	-	-	-	-	-	-
	US	1	1808	224	12.0	11.0-14.0	Inestimable	-	-
COVID-19 diagnosis	RT-PCR	2	1961	255	13.0	11.0-14.0	Inestimable	-	-
	Multiple modes	2	1365	179	12.0	10.0-14.0	Inestimable	-	-
Diabetes diagnosis	Method specified	2	1470	171	11.0	10.0-13.0	Inestimable	-	-
	Unclear	2	1856	263	13.0	12.0-15.0	Inestimable	-	-
DM type	Mixed type 1 and type 2 diabetes population	1	1317	140	11.0	9.0-12.0	Inestimable	-	-
	T2DM	2	1856	263	13.0	12.0-15.0	Inestimable	-	-
	Unclear	1	153	31	20.0	14.0-28.0	Inestimable	-	-
COVID-19 severity	Critically ill	1	48	39	81.0	67.0-91.0	Inestimable	-	-
	Not Critically ill	-	-	-	-	-	-	-	-
	Unclear	3	3278	395	13.0	10.0-16.0	Inestimable	-	-
Sample size	<100	1	48	39	81.0	67.0-91.0	Inestimable	-	-
	≥100	3	3278	395	13.0	10.0-16.0	Inestimable	-	-
Overall		4	3326	434	26.0	15.0-38.0	0.0-84.0	97.53	<0.01

Death in COVID-19 patients with diabetes who did not require ICU support

Subgroup	Category	Number of studies	Number of DM patients	Number of deaths	Mean prevalence of deaths		95% prediction interval	Heterogeneity measures	
					%	95% CI		I^2 (%)	Q (p-value)
Continent	Asia	12	743	131	11.0	5.0-20.0	0.0-48.0	*85.47*	<0.01
	Europe	3	5101	1531	35.0	25.0-45.0	Inestimable	-	-
	North America	3	2603	744	25.0	20.0-31.0	Inestimable	-	-
Country	China	7	566	105	12.0	3.0-23.0	0.0-58.0	*89.51*	<0.01
	France	-	-	-	-	-	--	-	-
	India	1	2	0	0.0	0.0-84.0	Inestimable	-	-
	Iran	1	113	11	10.0	5.0-17.0	Inestimable	-	-
	Italy	1	69	22	32.0	21.0-44.0	Inestimable	-	-
	Korea	2	42	11	21.0	9.0-35.0	Inestimable	-	-
	Oman	1	20	4	20.0	6.0-44.0	Inestimable	-	-
	UK	2	5036	1509	30.0	29.0-31.0	Inestimable	-	-
	US	3	2637	744	25.0	20.0-31.0	Inestimable	-	-
COVID-19 diagnosis	RT-PCR	15	3354	904	17.0	12.0-24.0	1.0-44.0	*89.16*	<0.01
	Multiple modes	3	5131	1502	23.0	11.0-38.0	Inestimable	-	-
Diabetes diagnosis	Method specified	3	238	60	23.0	6.0-46.0	Inestimable	-	-
	Unclear	15	8247	2346	19.0	14.0-23.0	6.0-36.0	*89.47*	<0.01
DM type	Mixed type 1 and type 2 diabetes population	2	156	62	40.0	32.0-47.0	Inestimable	-	-
	T2DM	2	7	0	0.0	0.0-41.0	Inestimable	-	-
	Unclear	15	8322	2344	17.0	13.0-22.0	4.0-35.0	*90.55*	<0.01

COVID-19 severity	Critically ill	2	16	4	19.0	2.0-44.0	Inestimable	-	-
	Not Critically ill	3	161	12	3.0	0.0-12.0	Inestimable	-	-
	Unclear	13	8308	2390	23.0	19.0-28.0	11.0-38.0	*87.77*	<0.01
Sample size	<100	11	414	101	16.0	6.0-29.0	0.0-66.0	*85.67*	<0.01
	≥100	7	8071	2305	22.0	17.0-26.0	8.0-39.0	*93.20*	<0.01
Overall		18	8485	2406	19.0	15.0-24.0	6.0-37.0	*89.72*	<0.01

Abbreviation: ICU: intensive care unit
*an ICU admission was determined for the number of inpatient diabetes patients treated in the ICU setting

Table S4. Univariate meta-regression analysis for the prevalence studies on diabetes mellitus among COVID-19 patients.[a,b]

Subgroup	Category	Univariate model[1]		
		OR	P value	95% CI
Country	China	1		
	UK	2.32	0.262	0.49-10.97
	US	1.67	0.367	0.51-5.53
	Korea	1.29	0.732	0.27-6.01
	Oman	2.44	0.374	0.30-19.82
	India	0.55	0.551	0.068-4.49
	Italy	1.06	0.951	0.13-8.62
	Iran	0.21	0.129	0.03-1.69
Continent	Asia	1		
	Europe	1.89	0.289	0.56-6.43
	North America	1.77	0.286	0.59-5.27
Study design	Case series	1		
	Case control	1.32	0.657	0.36-4.85
	Cross sectional	0.55	0.342	0.15-2.01
	Retrospective cohort	0.89	0.853	0.24-3.27
COVID-19 diagnosis method	RT-PCR	1		
	Multiple modes	0.73	0.558	0.25-2.17
COVID-19 severity	Unclear	1		
	Not critically ill	0.32	0.054*	0.10-1.02
	Critically ill	1.03	0.960	0.33-3.25
DM diagnosis method	Unclear	1		
	Method specified	1.83	0.247	0.64-5.24
DM types	Unclear	1		
	Mixed type 1 and type 2 diabetes population	1.84	0.387	0.43-7.85
	Type 2 DM	1.93	0.269	0.57-6.52
Sample size	<100	1		
	≥100	1.20	0.657	0.51-2.85

[1]statistically significant at p<0.1

[a] a multivariate metaregression analysis was not possible as only one variable in the univariate analysis showed statistical significance

[b] the analysis did not include the study based on diabetes patients only

*statistically significant finding

Table S5. Univariate and multivariate meta-regression analysis for the prevalence studies on deaths among COVID-19 patients with diabetes mellitus

Subgroup	Category	Univariate model[1]			Multivariate model[2a]		
		OR	P value	95% CI	OR	P value	95% CI
Country	China	1			-	-	-
	UK	1.70	0.557	0.24-12.05	-	-	-
	US	0.34	0.376	0.02-4.59	-	-	-
	Korea	0.70	0.613	0.15-3.23	-	-	-
	Oman	0.71	0.776	0.05-9.65	-	-	-
	India[#]	-	-	-	-	-	-
	Italy	1.33	0.812	0.10-18.06	-	-	-
	Iran	0.31	0.336	0.02-4.16	-	-	-
Continent	Asia	1			-	-	-
	Europe	1.16	0.805	0.32-4.23	-	-	-
	North America	0.76	0.659	0.21-2.77	-	-	-
Study design	Case series	1			-	-	-
	Case control	1.67	0.465	0.39-7.21	-	-	-
	Cross sectional	0.78	0.725	0.18-3.38	-	-	-
	Retrospective cohort	1.64	0.479	0.38-7.09	-	-	-
COVID-19 diagnosis method	RT-PCR	1			-	-	-
	Multiple modes	1.42	0.507	0.47-4.28	-	-	-
COVID-19 severity	Unclear	1			1		
	Not critically ill	0.42	0.278	0.08-2.16	0.50	0.383	0.09-2.64
	Critically ill	7.01	0.003*	2.13-23.04	5.52	0.014*	1.51-20.17
DM diagnosis method	Unclear	1			-	-	-
	Method specified	0.62	0.371	0.21-1.85	-	-	-
DM types	Unclear	1			-	-	-
	Mixed type 1 and type 2 diabetes population	1.37	0.620	0.37-5.10	-	-	-
	Type 2 DM	2.96	0.159	0.62-14.10	-	-	-
Sample size	<100	1			1		
	≥100	0.41	0.048*	0.17-0 .99	0.67	0.323	0.29-1.55

[1]Statistically significant at p<0.1; [2]Statistically significant at p<0.05
*Statistically significant finding
[a]model included: COVID-19 severity and sample size
[#]the number of deaths were 0 (zero)

Table S6. Univariate and multivariate meta-regression analysis for the prevalence studies on deaths among COVID-19 patients without diabetes mellitus.

Subgroup	Category	Univariate model[1]			Multivariate model[2a]			Multivariate model[2b]		
		OR	P value	95% CI	OR	P value	95% CI	OR	P value	95% CI
Country	China	1			1			-	-	-
	UK	4.62	0.078*	0.81-26.26	2.60	0.231	0.49-13.84	-	-	-
	US	1.67	0.417	0.44-6.43	2.03	0.225	0.60-6.86	-	-	-
	Korea	1.66	0.643	0.16-17.03	2.01	0.475	0.25-16.20	-	-	-
	Oman	0.27	0.244	0.026-2.79	0.33	0.263	0.04-2.66	-	-	-
	Italy	3.11	0.307	0.30-31.95	0.81	0.860	0.06-10.32	-	-	-
	Iran	0.99	0.994	0.10-10.20	0.26	0.264	0.02-3.29	-	-	-
Continent	Asia	1			-	-	-	1		
	Europe	4.36	0.028*	1.20-15.84	-	-	-	2.79	0.133	0.70-11.11
	North America	1.80	0.295	0.57-5.73	-	-	-	2.13	0.175	0.68-6.62
Study design	Case series	1			-	-	-	-	-	-
	Case control	1.10	0.910	0.21-5.57	-	-	-	-	-	-
	Cross sectional	0.91	0.900	0.18-4.63	-	-	-	-	-	-
	Retrospective cohort	1.18	0.827	0.23-6.04	-	-	-	-	-	-
COVID-19 diagnosis method	RT-PCR	1			1			1		
	Multiple modes	3.13	0.049*	1.01-9.73	4.63	0.068	0.87-24.62	2.50	0.139	0.71-8.77
COVID-19 severity	Unclear	1			-	-	-	-	-	-
	Not critically ill	0.31	0.291	0.03-3.07	-	-	-	-	-	-
	Critically ill	1.52	0.540	0.37-6.26	-	-	-	-	-	-
Sample size	<100	1			-	-	-	-	-	-
	≥100	0.89	0.823	0.31-2.59	-	-	-	-	-	-

[1]statistically significant at p<0.1; [2]statistically significant at p<0.05

*statistically significant finding

[a]model included: country of the hospitals, COVID-19 diagnosis method

[b]model included: continent of the hospitals, COVID-19 diagnosis method

Table S7. Quality assessment[3] of the reviewed studies*

Study design: Case series (n=13)

Study	1. Was the study question or objective clearly stated?	2. Was the study population clearly and fully described, including a case definition?	3. Were the cases consecutive ?	4. Were the subjects comparable?	6. Were the outcome measures clearly defined, valid, reliable, and implemented consistently across all study participants?	7. Was the length of follow-up adequate?	8. Were the statistical methods well-described?	9. Were the results well-described ?	Score (obtained /maximum possible)
	Questions used to assess the risk of bias in each study								
Cariou et al., 2020 [4]	1	1	0	1	0	1	1	1	81/104 (77.9%)
Bhandari et al., 2020 [5]	1	1	1	1	0	0	0	1	
Bhandari et al., 2020 [5]	1	1	0	1	0	1	1	1	
Ciceri et al., 2020 [6]	1	1	1	1	0	1	1	1	
Lee et al., 2020 [7]	1	1	0	1	0	1	1	1	
Zhao et al., 2020 [8]	1	1	1	1	0	1	1	1	
Marcello et al., 2020 [9]	1	1	0	1	0	1	1	1	
Rosenberg et al., 2020 [10]	1	1	0	1	0	1	1	1	
Wang et al., 2020 [11]	1	1	0	1	0	1	1	1	
Cen et al., 2020 [12]	1	1	1	1	0	1	1	1	
Jang et al., 2020 [2]	1	1	0	1	0	1	1	1	
Khamis et al., 2020 [13]	1	1	0	1	0	1	1	1	
Richardson et al., 2020 [14]	1	1	1	1	0	1	1	1	

Study design: Case control (n=3)

Study	1. Was the research question	2. Was the study popula	3. Did the auth	4. Were controls selected or recruited	5. Were the definitions, inclusion and exclusion criteria,	6. Were the cases	7. If less than 100 percent of eligible cases and/or	8. Was there use	9. Were the investigators able to confirm that the	10. Were the measures of exposure/risk clearly defined,	11. Were the assessor	12. Were key potential confounding variables measured	Score (obtained /maximu

	or objective in this paper clearly stated and appropriate?	tion clearly specified and defined?	ors include a sample size justification?	from the same or similar population that gave rise to the cases (including the same timeframe)?	algorithms or processes used to identify or select cases and controls valid, reliable, and implemented consistently across all study participants?	clearly defined and differentiated from controls?	controls were selected for the study, were the cases and/or controls randomly selected from those eligible?	of concurrent controls?	exposure/risk occurred prior to the development of the condition or event that defined a participant as a case?	valid, reliable, and implemented consistently (including the same time period) across all study participants?	s of exposure/risk blinded to the case or control status of participants?	and adjusted statistically in the analyses? If matching was used, did the investigators account for matching during study analysis?	m possible)
Yu et al., 2020 [15]	1	1	0	1	1	1	0	0	1	0	0	1	21/36 (58.3%)
Zhang et al., 2020 [16]	1	1	0	1	1	1	0	0	1	0	0	0	
Shi et al., 2020 [17]	1	1	0	1	1	1	1	1	0	0	0	1	

Study design: Cross-sectional (n=3)

Study	1. Was the research question or objective in this paper clearly stated?	2. Was the study population clearly specified and defined?	4. Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study prespecified and applied uniformly to all participants?	Was a sample size justification, power description, or variance and effect estimates provided?	6. For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?	7. Was the timeframe sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?	9. Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?	11. Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?	12. Were the outcome assessors blinded to the exposure status of participants?	14. Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?	Score (obtained /maximum possible)
Alkundi et al., 2020[18]	1	1	1	0	0	0	0	1	0	0	13/30 (43.33%)

Guan et al., 2020 [19]	1	1	1	1	0	0	0	0	0	1		
Nikpouraghdam et al., 2020 [20]	1	1	1	0	0	0	0	0	0	1		

Study design: Retrospective cohort (n=3)

Study	Was the research question or objective in this paper clearly stated?	Was the study population clearly specified and defined?	Was the participation rate of eligible persons at least 50%?	Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study prespecified and applied uniformly to all participants?	Was a sample size justification, power description, or variance and effect estimates provided?	For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?	7. Was the timeframe sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?	9. Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?	11. Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?	12. Were the outcome assessors blinded to the exposure status of participants?	13. Was loss to follow-up after baseline 20% or less?	14. Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?	Score (obtained /maximum possible)
Bode et al., 2020 [21]	1	1	1	1	0	1	0	0	0	0	1	0	20/36 (55.6%)
Zhang et al., 2020 [22]	1	1	1	1	0	1	0	0	0	0	1	1	
Yan et al., 2020 [23]	1	1	1	1	0	1	0	0	0	0	1	1	

Overall obtained/maximum score from 22 studies: 135/206 (65.5%)

[*] Risk of bias assessment done using the National Heart, Lung, and Blood Institute's tool.[3] For each risk of bias component, the answer of 'yes' was scored as 1 and otherwise scored as 0. Any such component that was not applicable to a particular study design has not been depicted in the table.

Reference

1. Liberati A, Altman DG, Tetzlaff J, Mulrow C, Gøtzsche PC, Ioannidis JPA, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. J Clin Epidemiol [Internet]. 2009 Oct;62(10):e1–34. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0895435609001802
2. Jang JG, Hur J, Choi EY, Hong KS, Lee W, Ahn JH. Prognostic Factors for Severe Coronavirus Disease 2019 in Daegu, Korea. J Korean Med Sci [Internet]. 2020 Jun;35(23):e209. Available from: https://jkms.org/DOIx.php?id=10.3346/jkms.2020.35.e209
3. Study Quality Assessment Tools | National Heart, Lung, and Blood Institute (NHLBI) [Internet]. [cited 2020 Jun 14]. Available from: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools
4. Cariou B, Hadjadj S, Wargny M, Pichelin M, Al-Salameh A, Allix I, et al. Phenotypic characteristics and prognosis of inpatients with COVID-19 and diabetes: the CORONADO study. Diabetologia [Internet]. 2020 Aug 29;63(8):1500–15. Available from: http://link.springer.com/10.1007/s00125-020-05180-x
5. Bhandari S, Bhargava A, Sharma S, Keshwani P, Sharma R, Banerjee S. Clinical Profile of Covid-19 Infected Patients Admitted in a Tertiary Care Hospital in North India. J Assoc Physicians India [Internet]. 2020 May;68(5):13–7. Available from: http://www.ncbi.nlm.nih.gov/pubmed/32610859
6. Ciceri F, Castagna A, Rovere-Querini P, De Cobelli F, Ruggeri A, Galli L, et al. Early predictors of clinical outcomes of COVID-19 outbreak in Milan, Italy. Clin Immunol [Internet]. 2020 Aug;217:108509. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1521661620304563
7. Lee JY, Kim HA, Huh K, Hyun M, Rhee J-Y, Jang S, et al. Risk Factors for Mortality and Respiratory Support in Elderly Patients Hospitalized with COVID-19 in Korea. J Korean Med Sci [Internet]. 2020 Jun;35(23):e223. Available from: https://jkms.org/DOIx.php?id=10.3346/jkms.2020.35.e223
8. Zhao J, Gao H-Y, Feng Z-Y, Wu Q-J. A Retrospective Analysis of the Clinical and Epidemiological Characteristics of COVID-19 Patients in Henan Provincial People's Hospital, Zhengzhou, China. Front Med [Internet]. 2020 Jun 5;7. Available from: https://www.frontiersin.org/article/10.3389/fmed.2020.00286/full
9. Marcello RK, Dolle J, Grami S, Adule R, Li Z, Tatem K, et al. Characteristics and Outcomes of COVID-19 Patients in New York City's Public Hospital System. medRxiv [Internet]. 2020 Jan 1;2020.05.29.20086645. Available from: http://medrxiv.org/content/early/2020/06/23/2020.05.29.20086645.abstract
10. Rosenberg ES, Dufort EM, Udo T, Wilberschied LA, Kumar J, Tesoriero J, et al. Association of Treatment With Hydroxychloroquine or Azithromycin With In-Hospital Mortality in Patients With COVID-19 in New York State. JAMA [Internet]. 2020 Jun 23;323(24):2493. Available from: https://jamanetwork.com/journals/jama/fullarticle/2766117
11. Wang X, Fang J, Zhu Y, Chen L, Ding F, Zhou R, et al. Clinical characteristics of non-critically ill patients with novel coronavirus infection (COVID-19) in a Fangcang Hospital. Clin Microbiol Infect [Internet]. 2020 Aug;26(8):1063–8. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1198743X20301774
12. Cen Y, Chen X, Shen Y, Zhang X-H, Lei Y, Xu C, et al. Risk factors for disease progression in patients with mild to moderate coronavirus disease 2019—a multi-centre observational study. Clin Microbiol Infect [Internet]. 2020 Sep;26(9):1242–7. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1198743X20303414
13. Khamis F, Al-Zakwani I, Al Naamani H, Al Lawati S, Pandak N, Omar MB, et al. Clinical characteristics and outcomes of the first 63 adult patients hospitalized with COVID-19: An experience from Oman. J Infect Public Health [Internet]. 2020 Jul;13(7):906–13. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1876034120304962
14. Richardson S, Hirsch JS, Narasimhan M, Crawford JM, McGinn T, Davidson KW, et al. Presenting Characteristics, Comorbidities, and Outcomes Among 5700 Patients Hospitalized With COVID-19 in the New York City Area. JAMA [Internet]. 2020 May 26;323(20):2052. Available from: https://jamanetwork.com/journals/jama/fullarticle/2765184
15. Yu C, Lei Q, Li W, Wang X, Liu W, Fan X, et al. Clinical Characteristics, Associated Factors, and Predicting COVID-19 Mortality Risk: A Retrospective Study in Wuhan, China. Am J Prev Med [Internet]. 2020 Aug;59(2):168–75. Available from: https://linkinghub.elsevier.com/retrieve/pii/S074937972030218X
16. Zhang N, Xu X, Zhou L-Y, Chen G, Li Y, Yin H, et al. Clinical characteristics and chest CT imaging features of critically ill COVID-19 patients. Eur Radiol [Internet]. 2020 May 30; Available from: http://link.springer.com/10.1007/s00330-020-06955-x
17. Shi Q, Zhang X, Jiang F, Zhang X, Hu N, Bimu C, et al. Clinical Characteristics and Risk Factors for Mortality of COVID-19 Patients With Diabetes in Wuhan, China: A Two-Center, Retrospective Study.

Diabetes Care [Internet]. 2020 Jul;43(7):1382–91. Available from: http://care.diabetesjournals.org/lookup/doi/10.2337/dc20-0598
18. Alkundi A, Mahmoud I, Musa A, Naveed S, Alshawwaf M. Clinical characteristics and outcomes of COVID-19 hospitalized patients with diabetes in the United Kingdom: A retrospective single centre study. Diabetes Res Clin Pract [Internet]. 2020 Jul;165:108263. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0168822720305131
19. Guan W-J, Liang W, Zhao Y, Liang H-R, Chen Z, Li Y, et al. Comorbidity and its impact on 1590 patients with COVID-19 in China: a nationwide analysis. Eur Respir J [Internet]. 2020 May;55(5):2000547. Available from: http://erj.ersjournals.com/lookup/doi/10.1183/13993003.00547-2020
20. Nikpouraghdam M, Jalali Farahani A, Alishiri G, Heydari S, Ebrahimnia M, Samadinia H, et al. Epidemiological characteristics of coronavirus disease 2019 (COVID-19) patients in IRAN: A single center study. J Clin Virol [Internet]. 2020 Jun;127:104378. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1386653220301207
21. Bode B, Garrett V, Messler J, McFarland R, Crowe J, Booth R, et al. Glycemic Characteristics and Clinical Outcomes of COVID-19 Patients Hospitalized in the United States. J Diabetes Sci Technol [Internet]. 2020 Jul 9;14(4):813–21. Available from: http://journals.sagepub.com/doi/10.1177/1932296820924469
22. Zhang Y, Cui Y, Shen M, Zhang J, Liu B, Dai M, et al. Association of diabetes mellitus with disease severity and prognosis in COVID-19: A retrospective cohort study. Diabetes Res Clin Pract [Internet]. 2020 Jul;165:108227. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0168822720304770
23. Yan Y, Yang Y, Wang F, Ren H, Zhang S, Shi X, et al. Clinical characteristics and outcomes of patients with severe covid-19 with diabetes. BMJ Open Diabetes Res Care [Internet]. 2020 Apr 27;8(1):e001343. Available from: http://drc.bmj.com/lookup/doi/10.1136/bmjdrc-2020-001343